

Terrestrial Non-Target Invertebrates and the Autumn Application of Synthetic Pyrethroids: Experimental Methodology and the Trade-off Between Replication and Plot Size

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Abstract. The effects of the synthetic pyrethroid, deltamethrin ((S)-*a*-cyano-3-phenoxybenzyl (1R)-cis-3-(2,2-dibromovinyl)-2,2-dimethylcyclopropanecarboxylate) on the non-target invertebrate fauna of winter-wheat was monitored over two cropping cycles. The product was applied at decimal growth stage 12–13 in the first week of November in 1986 and 1987. In 1986, demeton-S-methyl (*S*-2-ethylthioethyl *O,O*-dimethyl phosphorothioate) was applied as a toxic standard. Short-term depletions were detected in the autumn-active, field-overwintering Carabidae, especially *Bembidion obtusum*, *Nebria brevicollis* and *Pterostichus melanarius*. Staphylinidae were largely unaffected during the winter months however, significant increases in summer-active species were detected in both years. The most profound effects were on the Linyphiidae with 168–221 day depletions in the 4 ha plots in 1986/7 and 89–145 day reductions in the 1 ha plots used in 1987/8. *Meioneta rurestris* was the most affected species. In general, pyrethroid effects were greater than organophosphate effects, especially in the case of spiders. Summer aphid infestations tended to be lower in the pyrethroid-treated plots, possibly as a result of overwintering aphid control. No evidence for sub-lethal effects (number of eggs per female and proportion with empty crops) were found in the carabid, *Trechus quadristriatus* however, the proportion of female beetles was higher in the pyrethroid-treated plots. *B. obtusum* and *P. melanarius* numbers were significantly reduced within small, barriered areas 160 days after treatment within the pyrethroid-treated plots in 1987/8. This suggests that recolonisation of treated areas may reduce or dilute the duration of effects that would be detected in the whole-field treatments characteristic of the real world. Support for this was also obtained from the direct relationship between the duration of effects and plot size in the two years of the study. The potential for using models incorporating spatial dynamics to assist interpretation of experimental results is discussed.

throids to the beneficial invertebrate fauna of cereal fields. Synthetic pyrethroids have been approved for application against the aphid vectors of barley yellow dwarf virus (Scopes and Stables 1991) and this experimental question is relevant to predators of these aphids in the autumn and also the predators which overwinter in fields or their surrounds and prey upon summer aphid pests in the following year.

Previous studies have demonstrated effects of synthetic pyrethroids, applied in the autumn, on the beneficial fauna of cereals (Brown *et al.* 1988; Purvis *et al.* 1988). The effects reported were to a certain extent, predictable from the phenology of the species affected (Jepson 1989). For example, both studies detected statistically significant effects on autumn-breeding Carabidae (Coleoptera) that overwinter in fields or field boundaries (Sotherton 1984, 1985). The species affected included *Harpalus rufipes* (DeGeer), *Bembidion obtusum* Serville, *Nebria brevicollis* (F.) and *Trechus quadristriatus* (Schrank). Largely unaffected were the spring-breeding Carabidae that overwinter in field boundaries, including *B. lampros* (Herbst.). Where effects were detected upon Staphylinidae (Coleoptera), it was the winter-edaphic species that were affected (Good and Giller 1988) and not the summer or spring breeding guild of species. Linyphiidae (Arachnida) were generally also affected and are known to be particularly susceptible to this class of pesticides (Everts 1990).

Pesticide hazard to individual species in this diverse community of invertebrates is a function of a large number of interacting factors (Jepson 1988, 1989). These include the presence of a given species at the time of spraying, the use and availability of refugia, susceptibility, exposure and phenological variables, including dispersal rate and reproductive capacity. A semi-standardised methodology has evolved to measure short- to medium-term effects, *i.e.*, those occurring within a single cropping period. Generally, experiments take place within single fields, with application to replicated plots at the correct time with the recommended field rate of pesticide (Jepson 1992). The effects of single treatments of individual compounds are then measured, often in comparison to a toxic standard (Sotherton *et al.* 1987; Vickerman *et al.* 1987a, 1987b; Wratten *et al.* 1988). Population sampling methodology, plot sizes, replication, layout and statistical methods vary considerably between the published tests. All are assumed to provide a degree of

The underlying aim of this study was to determine experimentally, the hazard posed by autumn-applied, synthetic pyre-

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realism to standard agricultural practice such that hazard may be accurately predicted (Sotherton *et al.* 1988). This assumption has recently been challenged by the finding that experimental controls (which are usually untreated) may act as a source of invertebrates that invade treated plots and dilute the effects of treatment (Duffield and Baker 1990; Jepson and Thacker 1990; Thomas *et al.* 1990). The implications of this for experimental hazard analysis therefore require investigation.

We describe two experiments, carried out in successive years. In these we varied plot sizes, the number of treatments and the level of replication in an attempt to answer these more general questions, as well as determine pyrethroid hazard. A barrier treatment was also introduced in the second year to prevent reinvasion by some taxa.

Materials and Methods

The study was undertaken at Charity Down Farm of the Leckford Estates, near Stockbridge, Hampshire. The farm produces cereals for seed in a short rotation with herbage seed and peas. The experiments were carried out in 1986/7 and 1987/8. Treatments were applied in the first week of November to seedling winter-wheat plants at the three leaf stage (decimal growth stage (GS) 12–13; Zadoks *et al.* 1974).

The insecticide treatments were applied by farm staff using a tractor-mounted "Chafer Pathfinder" hydraulic sprayer with an 18-m boom fitted with Chafer "red cone" nozzles, 33 cm apart. The tractor travelled at 10 km/h and applied the pesticides at a volume rate of 220 l/ha with a tank pressure of 1.6 bar. In 1986, comparison was made between deltamethrin (2.5% EC) applied at 6.25 g/ai/ha, demeton-S-methyl (58% EC) applied at 243.6 g/ai/ha and an untreated control. The output from the nozzles was determined and the actual application rates were estimated to be 5.9 ± 0.1 and 228.5 ± 5.4 g/ai/ha, respectively. In 1987, deltamethrin alone was applied: the tractor calibration gave an estimated application rate of 6.0 ± 0.1 g/ai/ha.

The experimental layouts were different in the two years. In 1986, nine 4-ha plots (200 * 200 m) were marked out in two adjacent fields, partially separated by a hedgerow; treatments were assigned non-randomly to prevent the individual insecticide treatments being contiguous, thus artificially boosting the scale of any particular treatment. Relatively large plots were selected to limit the dilution effects of invertebrate diffusion from untreated areas: some replication was, however, included to meet the requirements for statistical analysis. For the purposes of analysis, the experimental design was treated as being within a single field since the two crops were in synchrony and the two fields were topographically similar.

In 1987, 10 1-ha plots (100 * 100 m) were marked out in a regular, alternating 2 * 5 array. Randomness in plot layout was sacrificed to ensure that each plot had the same neighbouring treatments: it was hoped that this would regularise interplot diffusion effects. The high level of replication of the single pyrethroid treatment was incorporated to permit statistical validation of 1986 results, at least for the period immediately after spraying before dispersal between plots could occur. It therefore permitted an investigation of the relationship between the duration of effects and the scale of treatment in the two years. In an attempt to determine the importance of interplot interference, each plot contained a 0.0049 ha (7 * 7 m) barriered sub-plot at its center. The barrier was constructed from 20 cm-wide strips of corrugated plastic lawn-edging material.

The epigeal invertebrate fauna was sampled, using pitfall traps (7 cm diameter plastic cups, inserted within a plastic drainpipe liner). These contained a 5% formalin solution (v/v) with a small quantity of detergent. In 1986, each plot contained a 5 * 4 array of traps, with a regular 10 m separation, at its center. In 1987, a 5 * 2 grid of traps was laid at the center of each plot. In both seasons, traps were emptied approximately every two weeks, for 38 and 26 weeks after application respec-

tively. Their contents were stored in 70% alcohol until identification of Carabidae, Staphylinidae and Linyphiidae was carried out to species level. In the spring of 1988, an additional five traps were set in each barriered sub-plot. These were emptied on five occasions between 13 and 28 April.

All the pitfall trap data were transformed by $\log(n + 1)$ and expressed as the number of individuals per trap per day for the purposes of analysis. An additional sampling method was used in 1987. From 31 May to 1 July, 'D-vac' vacuum insect net samples were taken in each plot on five occasions. Each sample was made up of five sub-samples, giving a total sample area of 0.46 m² per plot.

In 1987, the summer infestation of *Sitobion avenae* (F.) (Hemiptera: Aphididae) was recorded by visual inspection of 25 tillers per plot on several occasions between 30 May and 8 June. This was done to identify any indirect effects on aphid population dynamics resulting from natural enemy depletion. Aphids were identified to instar (Pullen 1990) however, only data for total infestation are given here.

Additionally in 1986/7, the gut contents and reproductive condition of *T. quadristriatus* was monitored to identify any sub-lethal effects that this species might be suffering. Up to 90 insects per treatment were examined on each sampling occasion. Beetles were sexed, the presence of mature eggs was determined in females and the gut contents of all individuals was classified as empty or full.

Results

Repeatable patterns of effect were apparent at the family level of taxonomic resolution. Short term reductions were apparent for deltamethrin in the Carabidae in both years (Figures 1, 2, Tables 1, 2) with significant depletions on one occasion in 1986/7 and on three successive occasions in 1987/8. In both years and with both products in 1986/7, there was a period of enhanced trapping rate after 50 days following treatment: this attenuated more rapidly in the 1-ha plots.

The species composition of the staphylinid samples was variable and the data were only expressed at the family level: the most abundant winter-active species was *Xantholinus linearis* (Olivier). No effects were detectable in the period following spraying in either year. Summer effects were however, found in 1987/8 where significant increases in summer-active species were found in the treated plots (Tables 1, 2).

There was evidence for longer-term depletions in both years for the Linyphiidae compared with the Carabidae and Staphylinidae (Figures 3, 4). Deltamethrin was apparently more toxic than demeton-S-methyl in 1986/7. Significant reductions were detected up to 168 days in the four ha plots and up to 89 days in the one ha plots although visual differences were detectable for at least 221 and 145 days, respectively.

The species were grouped into guilds of the important beneficial taxa. Data for the larvae of *N. brevicollis*, Acari, Collembola and Diptera are presented in Pullen (1990). The most affected Carabidae were those that breed in the autumn and overwinter in the field (Tables 1, 2). *B. obtusum* was reduced on three dates, immediately after spraying, in both years; in 1986/7 by both products. *N. brevicollis* and *P. melanarius* were both significantly reduced on single occasions in 1986/7, more than one month after spraying. Of the spring breeding, boundary overwintering guild, very few effects were detectable (Tables 1, 2). *N. biguttatus* was significantly reduced on one occasion, soon after spraying in 1987/8. *L. pilicornis* numbers were significantly enhanced, 160 days after treatment in the same year.

The spiders were the most affected group. Species within the Erigoninae and Linyphiinae were significantly reduced in both

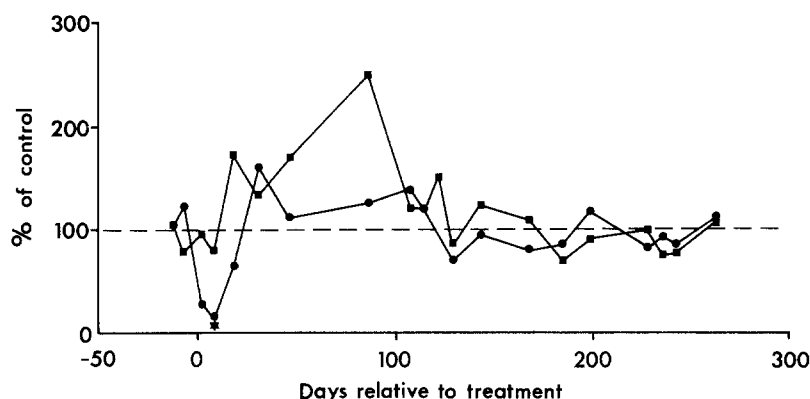


Fig. 1. Pitfall trap catches of total adult Carabidae in treated areas as a percentage of captures in control plots in 1986/7. Points labelled with an asterisk indicate a statistically significant difference with respect to control (One way ANOVA on $\log(x + 1)$ transformed data). Squares, demeton-S-methyl treatment; circles, deltamethrin treatment

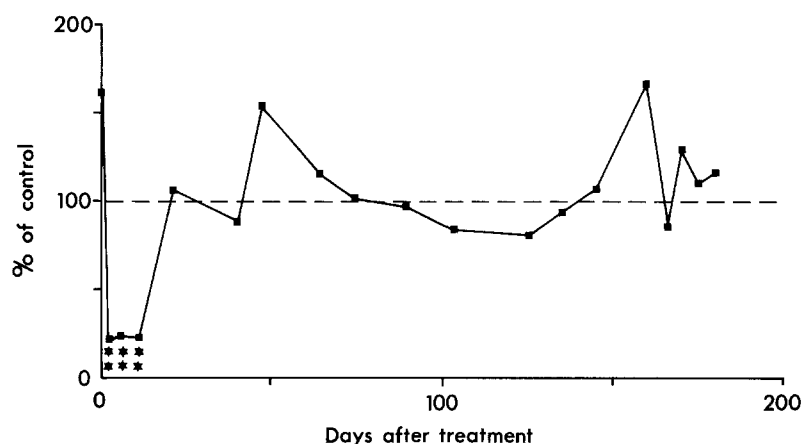


Fig. 2. Pitfall catches of total adult Carabidae in 1987/8. Asterisks indicate significant effects (see Figure 1 for details)

years on a range of occasions up to 185 days after treatment in 1986/7 and 175 days after treatment in 1987/8 (Tables 1, 2). *M. rurestris* was apparently the most affected species.

D-vac sampling of the non-target invertebrate fauna was undertaken between days 109 and 241 after treatment in 1986/7. No significant differences between treatments were found in the Carabidae, Staphylinidae or Linyphiidae (Pullen 1990). We were thus able to confirm the findings from the pitfall trap data and concluded that, despite the level of sampling being activity dependent, the pitfall traps were reflecting the true differences between treatments at this time of year. An infestation of *S. avenae* was recorded within the experiment in 1986/7 (Figures 5, 6). Total *S. avenae* were, in general, at lower levels in the autumn-treated plots and significantly so in the deltamethrin on one occasion (Figure 5). The proportion of tillers infested was equivalent in all plots (Figure 6). The trend of aphid infestation was reflected in the yields from each treatment: mean 1000 grain weights for the deltamethrin, demeton-S-methyl and control treatments (in grams $\pm$ 1 SE) were 53.51 ± 1.35 , 51.58 ± 0.45 and 50.27 ± 0.17 , respectively. These differences were too small to be significant (ANOVA, $P > 0.05$).

The *T. quadristriatus* study found no evidence for significant sub-lethal effects (Pullen 1990). The proportion of insects with full crops declined, in all treatments, from approximately 25% at the time of treatment, to approximately 10% after 200 days. The proportion of females in the sample was higher in the deltamethrin treatments by 11% on day 31, 30% on day 108 and 8% on day 122. The proportion of females with mature eggs fell, in all plots, from up to 70% just after treatment to between

20 and 30% after 200 days. There was some evidence for an early-season burst of egg production after 130 days in control and demeton-S-methyl treatments and not in the deltamethrin treatment; however, this effect was not significant. The mean number of eggs per female fell gradually from 2.5 at spraying to approximately 0.75 after 200 days.

Within the barriers (where reinvasion was prevented), over the period 160–180 days after treatment, invertebrate numbers tended to be consistently reduced in the treated plots; this reversed non-significant trends over the same period in the open field for *B. obtusum* and *P. melanarius* (Tables 1, 2). *B. obtusum* numbers were significantly reduced in the barrier treatments at 160 days (0.0834 ± 0.0406 and 0.2148 ± 0.0310 [$\log(n + 1)/\text{trap/day}$] in deltamethrin and control barriers respectively) and numerically less in the deltamethrin plots on all five sampling occasions. The same applied to *P. melanarius*, with fewer individuals trapped on four of the five sampling occasions in the treated plots ($P > 0.05$). The staphylinids were significantly reduced within the treated barriers, only at 160 days (0.0522 ± 0.0130 and 0.0981 ± 0.0312 [$\log(n + 1)/\text{trap/day}$] in deltamethrin and control barriers, respectively). They had identical numbers in both treatments on this date in the open field (Table 2). All the linyphiid species showed reductions in the treated barriers on all five sampling occasions, but significant effects were only detectable at the family level at 175 days (0.0986 ± 0.0213 and 0.1427 ± 0.0321 , data presented as above). These effects were similar to those detected in the field (Table 2).

To permit an examination of the relative levels of invertebrate depletion in the treated barriers and treated open field

Table 1. Mean (log x + 1) numbers ($\pm$ one standard error) of arthropods captured per trap per day pre- and post-treatment on 4 ha plots of winter wheat sprayed with solutions of aphicides (1 = deltamethrin, 2 = demeton-S-methyl) or 3 = remaining untreated, 1986–1987 Hampshire, southern England. F-ratio based on one-way ANOVA (2,6 DF). Mean numbers followed by the same letter do not differ significantly at the 5% level of significance

CARABIDAE		-12	-07	02	08	18	31	47	87	108	115	122
Autumn breeding/ field overwintering	<i>B. obesus</i>	1 .011 $\pm$.004	.118 $\pm$.033	.018 $\pm$.004a	.117 $\pm$ 0.49a	.057 $\pm$.024a	.856 $\pm$.137	.212 $\pm$.055	.014 $\pm$.004	.371 $\pm$.056	1.43 $\pm$.017	.443 $\pm$.052
	2 .015 $\pm$.005	.168 $\pm$.035	.827 $\pm$.068b	.646 $\pm$.093b	.203 $\pm$.022b	.203 $\pm$.022b	.829 $\pm$.068	.331 $\pm$.106	.171 $\pm$.152	.361 $\pm$.016	.153 $\pm$.025	.612 $\pm$.094
	3 .013 $\pm$.044	.086 $\pm$.018	.734 $\pm$.175b	.677 $\pm$.157b	.117 $\pm$.027ab	9.13*	.690 $\pm$.065	.200 $\pm$.065	.012 $\pm$.001	.318 $\pm$.059	.136 $\pm$.025	.480 $\pm$.081
	F-ratio			8.66*	8.30*							
	<i>N. brevicollis</i>	1 .116 $\pm$.006	.113 $\pm$.060	.054 $\pm$.006	.022 $\pm$.009	.011 $\pm$.004	.004 $\pm$.003	.006 $\pm$.000a	.000 $\pm$.---	.002 $\pm$.001	.000 $\pm$.---	.007 $\pm$.004
	2 .115 $\pm$.010	.074 $\pm$.055	.045 $\pm$.035	.026 $\pm$.008	.005 $\pm$.003	.005 $\pm$.003	.004 $\pm$.001	.005 $\pm$.000a	.000 $\pm$.---	.001 $\pm$.001	.000 $\pm$.---	.002 $\pm$.002
	3 .119 $\pm$.019	.168 $\pm$.117	.232 $\pm$.119	.104 $\pm$.064	.020 $\pm$.010	9.13*	.077 $\pm$.054	.039 $\pm$.013b	.001 $\pm$.001	.001 $\pm$.001	.001 $\pm$.001	.005 $\pm$.005
	F-ratio							7.07*				
	<i>P. melanarius</i>	1 .000 $\pm$.---	.003 $\pm$.001	.026 $\pm$.026	.000 $\pm$.---	.001 $\pm$.001	.001 $\pm$.001a	.001 $\pm$.001	.000 $\pm$.---	.001 $\pm$.000	.000 $\pm$.---	.003 $\pm$.000
	2 .000 $\pm$.---	.000 $\pm$.---	.014 $\pm$.014	.000 $\pm$.---	.002 $\pm$.002	.002 $\pm$.002	.001 $\pm$.001a	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.000	.000 $\pm$.---	.004 $\pm$.001
	3 .000 $\pm$.---	.003 $\pm$.003	.007 $\pm$.007	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.006 $\pm$.001b	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.000	.000 $\pm$.---	.001 $\pm$.001
	F-ratio						5.33*					
Spring breeding/ field overwintering	<i>T. quadrisratus</i>	1 .115 $\pm$.006	.444 $\pm$.208	.410 $\pm$.182	.137 $\pm$.061	.077 $\pm$.026	.522 $\pm$.129	.125 $\pm$.045	.005 $\pm$.002	.088 $\pm$.052	.030 $\pm$.017	.089 $\pm$.034
	2 .102 $\pm$.026	.240 $\pm$.191	.613 $\pm$.227	.271 $\pm$.069	.114 $\pm$.037	.114 $\pm$.037	.392 $\pm$.152	.100 $\pm$.077	.006 $\pm$.004	.046 $\pm$.039	.016 $\pm$.016	.062 $\pm$.035
	3 .097 $\pm$.026	.277 $\pm$.243	.586 $\pm$.218	.329 $\pm$.120	.064 $\pm$.037	.064 $\pm$.037	.365 $\pm$.145	.074 $\pm$.027	.003 $\pm$.002	.031 $\pm$.015	.015 $\pm$.013	.049 $\pm$.024
	F-ratio											
	<i>Amara</i> spp.	1 .002 $\pm$.001	.003 $\pm$.003	.007 $\pm$.007	.000 $\pm$.---	.001 $\pm$.001	.007 $\pm$.007	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.001 $\pm$.001	.000 $\pm$.---
	2 .003 $\pm$.002	.000 $\pm$.---	.014 $\pm$.014	.000 $\pm$.---	.001 $\pm$.001	.001 $\pm$.001	.000 $\pm$.---	.001 $\pm$.001	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---
	3 .003 $\pm$.002	.000 $\pm$.---	.007 $\pm$.007	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---
	F-ratio											
	<i>L. pilicornis</i>	1 .001 $\pm$.001	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---
	2 .003 $\pm$.002	.001 $\pm$.001	.000 $\pm$.---	.000 $\pm$.---	.004 $\pm$.001	.004 $\pm$.001	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---
	3 .003 $\pm$.002	.000 $\pm$.---	.014 $\pm$.007	.000 $\pm$.---	.004 $\pm$.003	.004 $\pm$.003	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---	.000 $\pm$.---
	F-ratio											
<i>N. biguttatus</i>	1 .007 $\pm$.003	.029 $\pm$.004	.042 $\pm$.022	.000 $\pm$.---	.004 $\pm$.004	.004 $\pm$.004	.025 $\pm$.008	.003 $\pm$.001	.000 $\pm$.---	.005 $\pm$.003	.000 $\pm$.---	.011 $\pm$.004
	2 .011 $\pm$.005	.028 $\pm$.006	.047 $\pm$.017	.000 $\pm$.---	.007 $\pm$.003	.007 $\pm$.003	.015 $\pm$.008	.003 $\pm$.001	.000 $\pm$.---	.002 $\pm$.001	.001 $\pm$.001	.006 $\pm$.003
	3 .005 $\pm$.001	.026 $\pm$.016	.060 $\pm$.011	.000 $\pm$.---	.005 $\pm$.001	.005 $\pm$.001	.027 $\pm$.001	.003 $\pm$.001	.000 $\pm$.000	.002 $\pm$.001	.000 $\pm$.---	.011 $\pm$.004

CARABIDAE			129	143	168	185	199	214	221	228	236	242	263
Autumn breeding/ field overwintering	<i>B. obesusum</i>	1	.308 ± .084	.163 ± .049	.225 ± .054	.048 ± .016	.225 ± .060	.085 ± .011	.052 ± .017	.112 ± .010	.075 ± .024	.084 ± .024	.067 ± .014
		2	.367 ± .102	.234 ± .043	.352 ± .044	.071 ± .015	.312 ± .065	.141 ± .060	.103 ± .021	.132 ± .021	.103 ± .022	.132 ± .025	.082 ± .007
		3	.422 ± .028	.181 ± .065	.296 ± .097	.093 ± .051	.363 ± .069	.167 ± .105	.098 ± .045	.214 ± .064	.207 ± .060	.135 ± .040	.091 ± .027
	<i>N. brevicollis</i>	1	.000 ± .---	.002 ± .001	.002 ± .001	.026 ± .012	.175 ± .099	.122 ± .025	.178 ± .034	.038 ± .004	.028 ± .009	.032 ± .010	.008 ± .004
		2	.000 ± .---	.002 ± .001	.004 ± .001	.020 ± .010	.097 ± .042	.046 ± .018	.154 ± .018	.036 ± .003	.028 ± .009	.021 ± .010	.010 ± .003
		3	.000 ± .---	.001 ± .001	.003 ± .002	.090 ± .030	.213 ± .058	.129 ± .018	.232 ± .015	.037 ± .008	.031 ± .006	.025 ± .011	.007 ± .005
	<i>P. melanarius</i>	1	.002 ± .002	.042 ± .005	.039 ± .005	.080 ± .009	.232 ± .055	.208 ± .003	.298 ± .007	.539 ± .057	.692 ± .048	.822 ± .053	.467 ± .007
		2	.000 ± .---	.049 ± .007	.042 ± .007	.055 ± .027	.178 ± .049	.209 ± .006	.302 ± .010	.578 ± .061	.521 ± .161	.726 ± .132	.447 ± .022
		3	.000 ± .---	.043 ± .013	.048 ± .011	.092 ± .007	.241 ± .028	.208 ± .009	.303 ± .013	.550 ± .047	.735 ± .024	.880 ± .013	.451 ± .028
	<i>T. quadristriatus</i>	1	.041 ± .015	.020 ± .006	.060 ± .027	.057 ± .029	.327 ± .167	.119 ± .060	.028 ± .044	.110 ± .055	.098 ± .051	.105 ± .024	.058 ± .014
		2	.026 ± .016	.007 ± .006	.020 ± .009	.016 ± .016	.119 ± .119	.055 ± .046	.176 ± .062	.180 ± .081	.043 ± .038	.127 ± .060	.056 ± .013
		3	.056 ± .032	.006 ± .003	.021 ± .010	.016 ± .014	.115 ± .109	.055 ± .035	.119 ± .050	.154 ± .064	.074 ± .028	.195 ± .069	.063 ± .018
Spring breeding/ field overwintering	<i>Amara</i> spp.	1	.000 ± .---	.000 ± .---	.005 ± .004	.003 ± .002	.012 ± .006	.003 ± .002	.026 ± .010	.022 ± .007	.016 ± .003	.028 ± .007	.007 ± .003
		2	.000 ± .---	.000 ± .---	.003 ± .002	.005 ± .003	.018 ± .010	.007 ± .003	.016 ± .013	.024 ± .006	.012 ± .005	.014 ± .008	.003 ± .001
		3	.000 ± .---	.000 ± .---	.002 ± .001	.004 ± .003	.004 ± .002	.005 ± .003	.014 ± .011	.016 ± .005	.014 ± .006	.023 ± .008	.007 ± .002
	<i>L. plicicornis</i>	1	.000 ± .---	.016 ± .006	.020 ± .007	.023 ± .008	.070 ± .014	.040 ± .008	.076 ± .006	.017 ± .028	.076 ± .013	.058 ± .022	.022 ± .001
		2	.000 ± .---	.016 ± .008	.020 ± .011	.012 ± .009	.078 ± .029	.042 ± .010	.086 ± .015	.054 ± .017	.088 ± .009	.086 ± .024	.025 ± .003
		3	.000 ± .---	.012 ± .015	.016 ± .010	.032 ± .013	.062 ± .033	.042 ± .006	.080 ± .014	.090 ± .020	.074 ± .012	.075 ± .012	.024 ± .003
	<i>N. biguttatus</i>	1	.006 ± .004	.029 ± .010	.015 ± .005	.013 ± .004	.040 ± .004	.007 ± .001	.028 ± .010	.024 ± .009	.041 ± .010	.036 ± .008	.010 ± .003
		2	.000 ± .---	.030 ± .009	.011 ± .003	.008 ± .003	.030 ± .011	.009 ± .004	.024 ± .012	.018 ± .009	.028 ± .004	.046 ± .006	.012 ± .002
		3	.000 ± .---	.031 ± .009	.018 ± .003	.016 ± .006	.026 ± .004	.005 ± .003	.024 ± .010	.032 ± .008	.035 ± .010	.057 ± .002	.014 ± .002
	<i>A. dorsale</i>	1	.000 ± .---	.000 ± .---	.002 ± .001	.016 ± .004	.134 ± .044	.067 ± .018	.041 ± .011	.026 ± .011	.056 ± .010	.058 ± .013	.047 ± .031
		2	.000 ± .---	.000 ± .---	.001 ± .001	.018 ± .007	.180 ± .059	.093 ± .021	.054 ± .016	.054 ± .027	.105 ± .019	.043 ± .015	.015 ± .005
		3	.000 ± .---	.000 ± .---	.000 ± .---	.024 ± .016	.120 ± .067	.072 ± .026	.087 ± .013	.080 ± .017	.080 ± .012	.050 ± .015	.011 ± .004
Spring breeding boundary overwintering	<i>B. lampros</i>	1	.000 ± .---	.002 ± .002	.050 ± .024	.147 ± .036	.302 ± .118	.160 ± .054	.264 ± .059	.312 ± .076	.361 ± .077	.336 ± .067	.224 ± .051
		2	.000 ± .---	.000 ± .---	.050 ± .030	.141 ± .039	.296 ± .107	.161 ± .065	.257 ± .068	.310 ± .071	.339 ± .077	.396 ± .059	.216 ± .057
		3	.000 ± .---	.000 ± .---	.060 ± .035	.120 ± .051	.261 ± .052	.114 ± .012	.236 ± .088	.295 ± .092	.287 ± .100	.283 ± .102	.181 ± .071

Table 1 (continued). Mean (log x + 1) numbers ($\pm$ one standard error) of arthropods captured per trap per day pre- and post-treatment on 4 ha plots of winter wheat sprayed with solutions of aphicides (1 = deltamethrin, 2 = demeton-S-methyl) or 3 = remaining untreated, 1986–1987 Hampshire, southern England. F-ratio based on one-way ANOVA (2,6 DF). Mean numbers followed by the same letter do not differ significantly at the 5% level of significance

STAPHYLINIDAE	1	.006 $\pm$.004	.018 $\pm$.003	.006 $\pm$.002	.082 $\pm$.021	.346 $\pm$.058	.351 $\pm$.054	.517 $\pm$.017	.179 $\pm$.037	.225 $\pm$.021	.220 $\pm$.036	.080 $\pm$.027
	2	.010 $\pm$.005	.015 $\pm$.003	.009 $\pm$.003	.108 $\pm$.039	.321 $\pm$.064	.332 $\pm$.030	.539 $\pm$.026	.136 $\pm$.036	.193 $\pm$.046	.201 $\pm$.052	.074 $\pm$.014
	3	.020 $\pm$.008	.020 $\pm$.007	.015 $\pm$.005	.122 $\pm$.161	.339 $\pm$.101	.360 $\pm$.017	.416 $\pm$.040	.212 $\pm$.020	.166 $\pm$.050	.203 $\pm$.018	.112 $\pm$.005
ERIGONINAE												
<i>E. ara</i>	1	.004 $\pm$.002a	.023 $\pm$.004	.030 $\pm$.002	.093 $\pm$.013	.270 $\pm$.032	.179 $\pm$.050	.178 $\pm$.017	.179 $\pm$.049	.151 $\pm$.030	.280 $\pm$.032	.185 $\pm$.054
	2	.078 $\pm$.021a	.087 $\pm$.016	.123 $\pm$.019	.166 $\pm$.054	.337 $\pm$.041	.175 $\pm$.007	.231 $\pm$.051	.217 $\pm$.029	.218 $\pm$.067	.429 $\pm$.122	.179 $\pm$.068
	3	.147 $\pm$.048b	.122 $\pm$.047	.163 $\pm$.052	.218 $\pm$.064	.397 $\pm$.041	.245 $\pm$.027	.183 $\pm$.034	.182 $\pm$.032	.283 $\pm$.074	.533 $\pm$.118	.291 $\pm$.034
F-ratio		5.51*										
<i>E. denipalpis</i>	1	.000 $\pm$.---	.002 $\pm$.001a	.002 $\pm$.001	.008 $\pm$.002	.028 $\pm$.005	.017 $\pm$.004	.054 $\pm$.004	.030 $\pm$.007	.016 $\pm$.003	.065 $\pm$.030	.018 $\pm$.002
	2	.010 $\pm$.005	.007 $\pm$.001b	.014 $\pm$.001	.021 $\pm$.008	.087 $\pm$.033	.038 $\pm$.013	.043 $\pm$.005	.039 $\pm$.008	.043 $\pm$.013	.018 $\pm$.041	.011 $\pm$.004
	3	.014 $\pm$.002	.004 $\pm$.001ab	.010 $\pm$.006	.021 $\pm$.008	.091 $\pm$.016	.038 $\pm$.012	.041 $\pm$.009	.034 $\pm$.010	.043 $\pm$.012	.111 $\pm$.036	.019 $\pm$.002
F-ratio			6.38*									
<i>Oedothorax</i> spp.	1	.024 $\pm$.010	.009 $\pm$.002	.018 $\pm$.001a	.029 $\pm$.009	.089 $\pm$.019	.051 $\pm$.010	.047 $\pm$.032	.070 $\pm$.006	.041 $\pm$.010	.201 $\pm$.057	.058 $\pm$.017
	2	.008 $\pm$.002	.022 $\pm$.007	.036 $\pm$.004b	.063 $\pm$.016	.147 $\pm$.028	.050 $\pm$.008	.058 $\pm$.026	.068 $\pm$.019	.015 $\pm$.020	.366 $\pm$.039	.074 $\pm$.025
	3	.006 $\pm$.004	.016 $\pm$.004	.024 $\pm$.002a	.059 $\pm$.020	.171 $\pm$.031	.071 $\pm$.011	.070 $\pm$.009	.082 $\pm$.014	.100 $\pm$.026	.334 $\pm$.068	.085 $\pm$.019
F-ratio				18.53***								
LINYPHIINAE												
<i>B. gracilis</i>	1	.004 $\pm$.002a	.006 $\pm$.002	.003 $\pm$.001	.008 $\pm$.001a	.035 $\pm$.012	.014 $\pm$.005	.028 $\pm$.008	.020 $\pm$.004	.021 $\pm$.006	.046 $\pm$.010	.013 $\pm$.002
	2	.006 $\pm$.000ab	.003 $\pm$.002	.003 $\pm$.001	.008 $\pm$.001a	.024 $\pm$.009	.012 $\pm$.005	.022 $\pm$.004	.022 $\pm$.007	.014 $\pm$.005	.071 $\pm$.004	.017 $\pm$.002
	3	.022 $\pm$.005b	.001 $\pm$.001	.001 $\pm$.001	.018 $\pm$.003b	.039 $\pm$.017	.012 $\pm$.005	.032 $\pm$.002	.020 $\pm$.005	.014 $\pm$.006	.058 $\pm$.015	.014 $\pm$.001
F-ratio		9.15*			10.28*							
<i>L. tenels</i>	1	.014 $\pm$.007	.007 $\pm$.003	.001 $\pm$.001	.008 $\pm$.004	.012 $\pm$.000	.013 $\pm$.001	.020 $\pm$.009	.030 $\pm$.003	.028 $\pm$.001	.067 $\pm$.010	.010 $\pm$.003
	2	.004 $\pm$.002	.005 $\pm$.001	.002 $\pm$.001	.009 $\pm$.005	.022 $\pm$.011	.009 $\pm$.002	.026 $\pm$.002	.030 $\pm$.003	.014 $\pm$.006	.075 $\pm$.017	.015 $\pm$.003
	3	.014 $\pm$.005	.010 $\pm$.004	.006 $\pm$.004	.013 $\pm$.002	.026 $\pm$.005	.010 $\pm$.004	.034 $\pm$.005	.024 $\pm$.007	.028 $\pm$.003	.076 $\pm$.021	.015 $\pm$.005
F-ratio												
<i>M. rurescens</i>	1	.000 $\pm$.---	.002 $\pm$.001	.009 $\pm$.002a	.018 $\pm$.005	.035 $\pm$.020	.023 $\pm$.006	.081 $\pm$.012	.063 $\pm$.031	.016 $\pm$.006	.019 $\pm$.005	.006 $\pm$.002
	2	.010 $\pm$.002b	.009 $\pm$.000	.031 $\pm$.005b	.036 $\pm$.014	.055 $\pm$.022	.022 $\pm$.007	.070 $\pm$.019	.083 $\pm$.021	.016 $\pm$.002	.032 $\pm$.009	.019 $\pm$.009
	3	.020 $\pm$.002c	.022 $\pm$.012	.022 $\pm$.003ab	.054 $\pm$.006	.077 $\pm$.007	.036 $\pm$.008	.069 $\pm$.009	.076 $\pm$.017	.011 $\pm$.003	.050 $\pm$.008	.019 $\pm$.011
F-ratio		38.77***		10.77*								

Table 2. Mean ($\log x + 1$) numbers ($\pm$ one standard error) of arthropods captured per trap per day pre- and post-treatment on 1 ha plots of winter wheat sprayed with solution of 1 = deltamethrin or 2 = remaining untreated, 1987–1988 Hampshire, southern England.

CARABIDAE		0	02	06	11	21	40	47	64	74	
Autumn feeding/field overwintering	<i>B. obesus</i>	1	.669 ± .066	.083 ± .016	.064 ± .011	.930 ± .015	.478 ± .033	.998 ± .108	.721 ± .047	.778 ± .032	
	t-value	2	.539 ± .036 -9.18**	.383 ± .016 -5.59**	.314 ± .028 -8.27**	.903 ± .053	.534 ± .034	.874 ± .072	.669 ± .061	.754 ± .075	
	<i>N. brevicollis</i>	1	.005 ± .005	.004 ± .004	.015 ± .008	.000 ± .---	.025 ± .004	.005 ± .002	.044 ± .019	.021 ± .008	.010 ± .003
Spring breeding/field overwintering	<i>P. melanarius</i>	2	.015 ± .003	.028 ± .015	.006 ± .006	.000 ± .---	.002 ± .001	.036 ± .011	.026 ± .010	.009 ± .004	
	t-value	1	.005 ± .003	.000 ± .---	.000 ± .---	.000 ± .---	.001 ± .000	.011 ± .003	.005 ± .000	.000 ± .---	
	<i>T. quadrisratus</i>	2	.002 ± .002	.000 ± .---	.009 ± .004	.000 ± .---	.087 ± .002	.000 ± .---	.009 ± .003	.005 ± .002	.000 ± .---
Spring breeding/boundary overwintering	<i>L. pilicornis</i>	1	.348 ± .053	.086 ± .012	.048 ± .017	.039 ± .014	.079 ± .013	.296 ± .089	.131 ± .031	.042 ± .008	
	t-value	2	.211 ± .003	.214 ± .043 -2.88*	.144 ± .007 -5.35**	.101 ± .015 -2.99*	.299 ± .032	.044 ± .016	.159 ± .038	.095 ± .020	.040 ± .008
	<i>N. biguttatus</i>	1	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .---
STAPHYLINIDAE	t-value	2	.001 ± .---	.001 ± .---	.000 ± .---	.029 ± .005	.005 ± .004	.017 ± .006	.013 ± .002	.000 ± .---	
	<i>B. lampros</i>	1	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .000	.000 ± .---	.000 ± .---	.000 ± .---	
	t-value	2	.000 ± .---	.000 ± .---	.000 ± .---	.001 ± .001	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .---	
ERIGONINAE	<i>E. atra</i>	1	.103 ± .023	.045 ± .011	.062 ± .009	.000 ± .---	.032 ± .006	.110 ± .026	.056 ± .016	.020 ± .009	
	t-value	2	.143 ± .025	.114 ± .027 -2.33*	.078 ± .014	.000 ± .---	.032 ± .015	.065 ± .026	.054 ± .011	.019 ± .005	
	<i>E. dentipalpis</i>	1	.007 ± .003	.000 ± .---	.000 ± .---	.000 ± .---	.008 ± .002	.007 ± .001	.017 ± .003	.002 ± .001	.003 ± .001
LINYPHIINAE	t-value	2	.007 ± .003	.000 ± .---	.004 ± .004	.000 ± .---	.007 ± .001	.021 ± .002	.011 ± .005	.009 ± .002	
	<i>Oedothorax</i> spp.	1	.002 ± .002	.000 ± .---	.000 ± .---	.000 ± .---	.001 ± .001	.000 ± .---	.000 ± .---	.000 ± .---	
	t-value	2	.003 ± .002	.000 ± .---	.000 ± .---	.000 ± .---	.005 ± .002	.003 ± .002	.006 ± .002	.002 ± .002	
<i>B. gracilis</i>	<i>Oedothorax</i> spp.	1	.014 ± .003	.000 ± .---	.011 ± .005	.000 ± .---	.004 ± .002	.009 ± .001	.003 ± .001	.003 ± .002	
	t-value	2	.020 ± .006	.004 ± .004	.000 ± .---	.000 ± .---	.007 ± .001	.003 ± .001	.008 ± .003	.002 ± .002	
	<i>L. tenuis</i>	1	.002 ± .002	.000 ± .---	.000 ± .---	.000 ± .---	.012 ± .003	.007 ± .001	.007 ± .001	.003 ± .001	.000 ± .---
<i>M. rurescens</i>	t-value	2	.000 ± .---	.000 ± .---	.000 ± .---	.000 ± .---	.009 ± .003	.004 ± .001	.005 ± .001	.001 ± .001	
	<i>L. tenuis</i>	1	.020 ± .004	.004 ± .004	.009 ± .004	.000 ± .---	.011 ± .002	.018 ± .008	.004 ± .001	.000 ± .---	
	t-value	2	.015 ± .004	.021 ± .009	.006 ± .006	.000 ± .---	.037 ± .003 -6.00**	.014 ± .001	.013 ± .006	.007 ± .001	.003 ± .001
	<i>M. rurescens</i>	1	.022 ± .010	.004 ± .004	.002 ± .002	.000 ± .---	.001 ± .001	.006 ± .004	.002 ± .001	.003 ± .002	
	t-value	2	.020 ± .006	.029 ± .006 -3.84**	.000 ± .---	.000 ± .---	.003 ± .001	.005 ± .002	.005 ± .002	.007 ± .002	

Table 2 (continued). Mean (log x + 1) numbers ($\pm$ one standard error) of arthropods captured per trap per day pre- and post-treatment on 1 ha plots of winter wheat sprayed with solution of 1 = deltamethrin or 2 = remaining untreated, 1987–1988 Hampshire, southern England

CARABIDAE		89	103	125	135	145	160	166	170	175	180
Autumn feeding/ field overwintering	<i>B. obsoletus</i>	1	.478 $\pm$.031	.170 $\pm$.023	.286 $\pm$.057	.299 $\pm$.021	.060 $\pm$.005	.207 $\pm$.021	.225 $\pm$.028	.210 $\pm$.031	.061 $\pm$.011
	t-value	2	.474 $\pm$.058	.199 $\pm$.030	.127 $\pm$.102	.304 $\pm$.036	.063 $\pm$.005	.136 $\pm$.031	.278 $\pm$.047	.151 $\pm$.023	.056 $\pm$.011
	<i>N. brevicollis</i>	1	.008 $\pm$.001	.001 $\pm$.001	.001 $\pm$.001	.000 $\pm$.001	.002 $\pm$.001	.002 $\pm$.002	.001 $\pm$.001	.006 $\pm$.003	.002 $\pm$.002
	t-value	2	.002 $\pm$.002	.000 $\pm$.001	.009 $\pm$.007	.000 $\pm$.001	.001 $\pm$.001	.007 $\pm$.002	.000 $\pm$.001	.002 $\pm$.002	.000 $\pm$.001
	<i>P. melanarius</i>	1	.000 $\pm$.001	.000 $\pm$.001	.003 $\pm$.001	.007 $\pm$.001	.031 $\pm$.003	.068 $\pm$.012	.064 $\pm$.013	.041 $\pm$.012	.012 $\pm$.003
	t-value	2	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.000	.006 $\pm$.001	.022 $\pm$.004	.030 $\pm$.009	.069 $\pm$.008	.064 $\pm$.023	.005 $\pm$.003
	<i>T. quadricollis</i>	1	.021 $\pm$.001	.006 $\pm$.002	.019 $\pm$.004	.018 $\pm$.004	.003 $\pm$.002	.012 $\pm$.007	.013 $\pm$.007	.013 $\pm$.005	.003 $\pm$.002
	t-value	2	.023 $\pm$.005	.003 $\pm$.001	.009 $\pm$.003	.030 $\pm$.005	.003 $\pm$.001	.000 $\pm$.001	.007 $\pm$.002	.002 $\pm$.002	.000 $\pm$.001
	<i>L. pilicornis</i>	1	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.028 $\pm$.002	.031 $\pm$.006	.052 $\pm$.006	.038 $\pm$.010	.043 $\pm$.011	.044 $\pm$.006
	t-value	2	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.027 $\pm$.002	.032 $\pm$.005	.015 $\pm$.006	.031 $\pm$.014	.057 $\pm$.018	.013 $\pm$.003
Spring breeding/ boundary overwintering	<i>N. biguttatus</i>	1	.005 $\pm$.001	.000 $\pm$.001	.002 $\pm$.001	.007 $\pm$.002	.134 $\pm$.003	.017 $\pm$.007	.006 $\pm$.003	.021 $\pm$.010	.009 $\pm$.004
	t-value	2	.008 $\pm$.001	.000 $\pm$.001	.002 $\pm$.002	.007 $\pm$.002	.016 $\pm$.003	.023 $\pm$.008	.009 $\pm$.003	.015 $\pm$.005	.010 $\pm$.002
	<i>B. lampros</i>	1	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.025 $\pm$.008	.048 $\pm$.013	.063 $\pm$.015	.133 $\pm$.002	.070 $\pm$.001
	t-value	2	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.019 $\pm$.006	.050 $\pm$.015	.050 $\pm$.019	.080 $\pm$.013	.075 $\pm$.007
STAPHYLINIDAE	<i>E. atra</i>	1	.008 $\pm$.001	.000 $\pm$.001	.006 $\pm$.001	.020 $\pm$.002	.004 $\pm$.001	.033 $\pm$.008	.011 $\pm$.004	.120 $\pm$.021	.044 $\pm$.013
	t-value	2	.010 $\pm$.002	.000 $\pm$.001	.011 $\pm$.002	.020 $\pm$.003	.003 $\pm$.003	.033 $\pm$.008	.014 $\pm$.006	.029 $\pm$.007	.024 $\pm$.003
	<i>E. denigralis</i>	1	.004 $\pm$.001	.005 $\pm$.003	.010 $\pm$.002	.020 $\pm$.003	.139 $\pm$.012	.168 $\pm$.017	.104 $\pm$.027	.173 $\pm$.018	.138 $\pm$.014
	t-value	2	.009 $\pm$.001	.004 $\pm$.002	.043 $\pm$.012	.030 $\pm$.003	.145 $\pm$.011	.150 $\pm$.031	.121 $\pm$.026	.133 $\pm$.021	.129 $\pm$.012
ERYTHRIDAE	<i>E. denigralis</i>	1	.000 $\pm$.001	.000 $\pm$.001	.001 $\pm$.001	.010 $\pm$.002	.034 $\pm$.006	.033 $\pm$.004	.028 $\pm$.003	.020 $\pm$.006	.000 $\pm$.001
	t-value	2	.003 $\pm$.001	.000 $\pm$.001	.002 $\pm$.001	.010 $\pm$.002	.026 $\pm$.003	.022 $\pm$.008	.030 $\pm$.014	.043 $\pm$.010	.015 $\pm$.003
	<i>Oedothorax</i> spp.	1	.003 $\pm$.001	.000 $\pm$.001	.002 $\pm$.001	.002 $\pm$.001	.009 $\pm$.002	.022 $\pm$.004	.011 $\pm$.006	.012 $\pm$.006	.009 $\pm$.003
	t-value	2	.003 $\pm$.001	.000 $\pm$.001	.001 $\pm$.001	.002 $\pm$.001	.009 $\pm$.002	.022 $\pm$.009	.017 $\pm$.003	.021 $\pm$.003	.007 $\pm$.003
LINYPHINAE	<i>B. gracilis</i>	1	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.003 $\pm$.001	.009 $\pm$.004	.004 $\pm$.003	.004 $\pm$.003	.006 $\pm$.003
	t-value	2	.000 $\pm$.001	.000 $\pm$.001	.001 $\pm$.001	.000 $\pm$.001	.003 $\pm$.001	.002 $\pm$.002	.000 $\pm$.001	.000 $\pm$.001	.003 $\pm$.002
	<i>L. tenuis</i>	1	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.000	.000 $\pm$.000	.003 $\pm$.003	.005 $\pm$.002	.001 $\pm$.001	.002 $\pm$.002	.003 $\pm$.003
	t-value	2	.003 $\pm$.001	.000 $\pm$.001	.002 $\pm$.001	.000 $\pm$.001	.002 $\pm$.001	.002 $\pm$.002	.003 $\pm$.002	.002 $\pm$.002	.005 $\pm$.003
<i>M. rurestris</i>	<i>M. rurestris</i>	1	.003 $\pm$.001	.000 $\pm$.001	.005 $\pm$.002	.005 $\pm$.002	.027 $\pm$.004	.031 $\pm$.012	.020 $\pm$.006	.035 $\pm$.008	.014 $\pm$.002
	t-value	2	.005 $\pm$.001	.000 $\pm$.001	.004 $\pm$.003	.004 $\pm$.000	.036 $\pm$.007	.047 $\pm$.001	0.36 $\pm$.002	.064 $\pm$.016	.030 $\pm$.006
	<i>B. lampros</i>	1	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.025 $\pm$.008	.048 $\pm$.013	.063 $\pm$.015	.133 $\pm$.002	.070 $\pm$.001
	t-value	2	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.000 $\pm$.001	.019 $\pm$.006	.050 $\pm$.015	.050 $\pm$.019	.080 $\pm$.013	.075 $\pm$.007

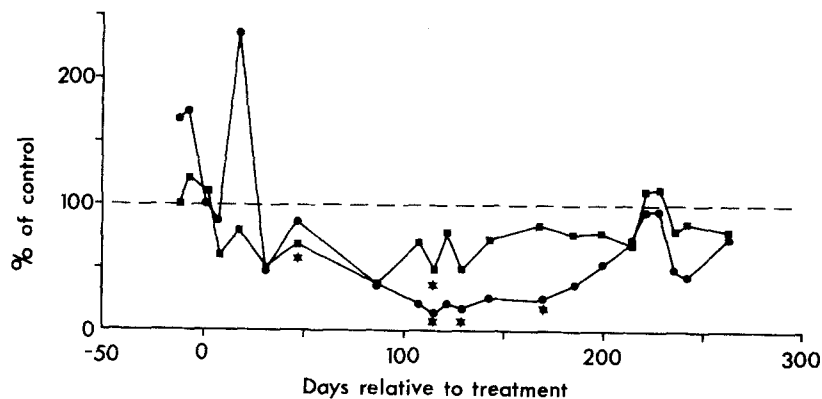


Fig. 3. Pitfall catches of total adult Linyphiidae in 1986/7 (see Figure 1 for details)

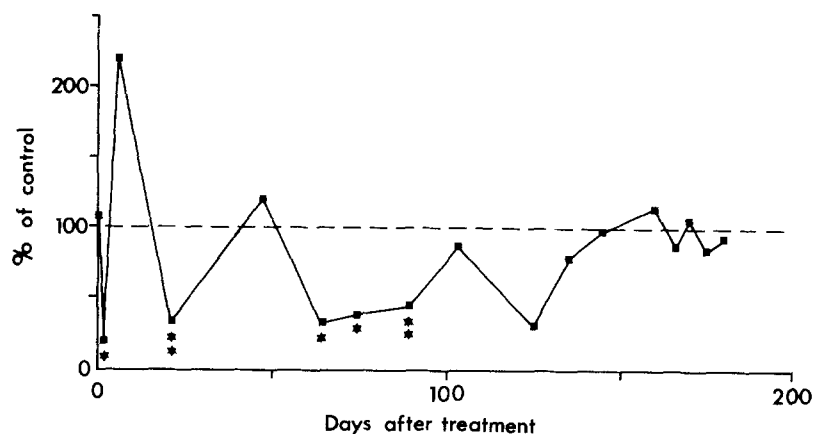


Fig. 4. Pitfall catches of total adult Linyphiidae in 1987/8 (see Figure 1 for details)

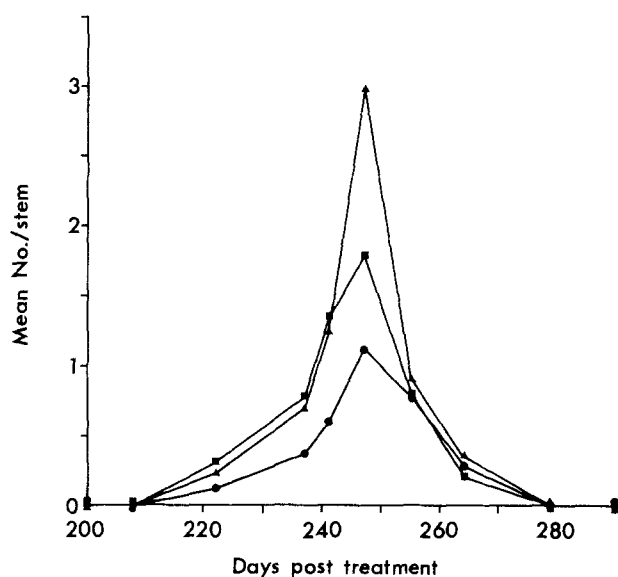


Fig. 5. Total numbers of *S. avenae* per stem in the summer of 1987. Twenty stems sampled randomly per plot. Deltamethrin treatment significantly lower than other two on fifth sampling occasion. Closed circles deltamethrin; squares, demeton-S-methyl; triangles, control

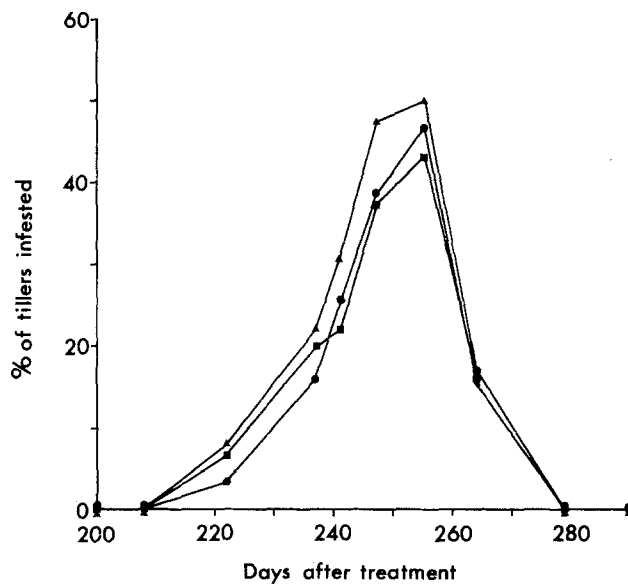


Fig. 6. Percentage of stems infested with *S. avenae* in the summer of 1987

plots the percentage differences between treated and control areas for the two sampling zones are presented in Figure 7. In all cases, in the groups selected, barrier populations were affected more severely after this long period post-spraying; the staphylinids showing the greatest effects.

Discussion

At the family level, there was evidence for a significant short-term depletion of certain Carabidae, following exposure to deltamethrin. There was also evidence for an enhancement of trapping rate for a period, at least 60 days following pyrethroid

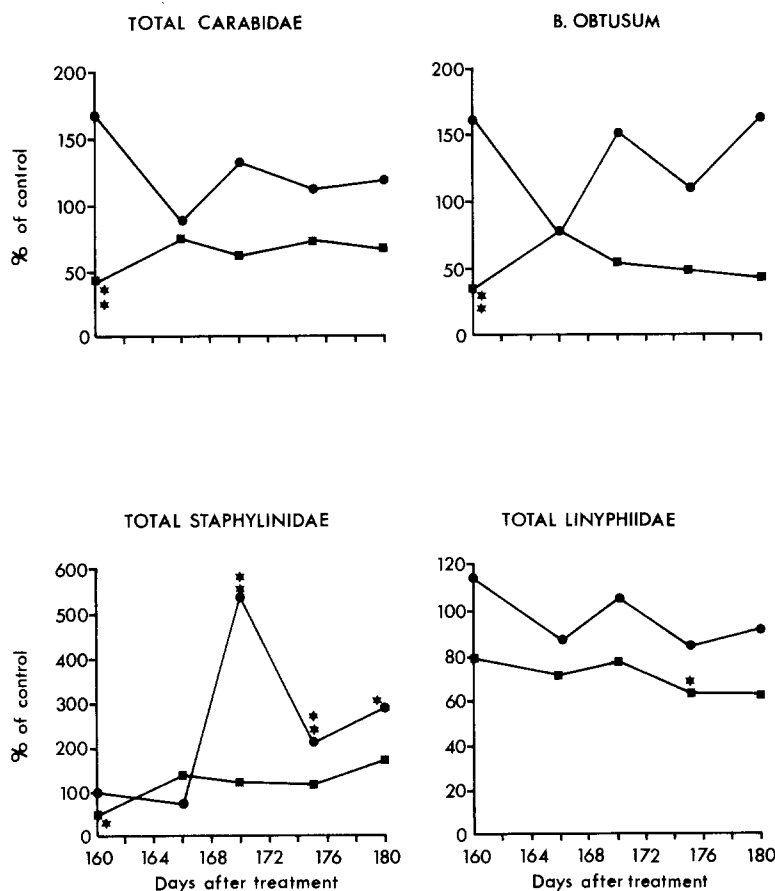


Fig. 7. Trapping rates of different taxa in deltamethrin-treated open plots and barred areas in 1987/8 experiment. Asterisks indicate statistically significant difference from control (*, $p < 0.05$; **, $p < 0.01$; t-test) for actual values, see text. Circles, open field; squares, barred plots

treatment in both years. This cannot be explained however, there may have been a degree of enhanced activity in the treated plots leading to a higher probability of capture. The causes of this are likely to have been indirect (food depletion) after such a long interval had elapsed, following spraying.

The Staphylinidae were, in general, not significantly affected, although there was a significant increase in trapping rate during the summer period in 1987/8 in unbarriered areas. Again, no explanation for this effect can be offered and verification is required.

Linyphiidae were the most affected group, with deltamethrin having significantly more severe effects than demeton-S-methyl. This family is known to be susceptible to synthetic pyrethroids (*e.g.*, Everts 1990). In both years a period of enhanced trapping, probably a result of behavioural activation, was detected, following treatment. This is again likely to be a symptom of the poisoning process although it is unknown if all sensitised individuals eventually succumb to the pesticide.

Overall, demeton-S-methyl was less toxic than deltamethrin and may constitute an environmentally softer compound for pest management purposes on the basis of reduced non-target invertebrate side effects. The importance of the effects that were detected are difficult to gauge and must partly be a function of the currency of effect under consideration. Autumn/winter active beneficial invertebrates are known to feed upon aphids (Sopp and Chiverton 1987), although the economic benefit of this is unknown. Linyphiidae are known to be predators of aphids (Sunderland *et al.* 1986) and the reduced spider densities were potentially important in terms of both winter and spring aphid predation, although this was confounded with

direct pesticide effects upon aphids in the present study. The importance of the relatively transitory effects for the long-term population dynamics of some groups cannot be determined, partly because these effects will have been a function of scale of treatment and the presence of unsprayed reservoirs within the field (Jepson and Thacker 1990; Thomas *et al.* 1990) and also because more subtle effects on resource gathering and overwintering survival were not recorded. It is therefore, not yet possible to confirm Inglesfield's (1989) assertion that these perturbations have no long-term significance for the groups concerned.

As far as effects on individual species was concerned, it would seem that most taxa active in the field at the time of spraying, were affected. In particular, the autumn breeding, field overwintering Carabidae, winter edaphic Staphylinidae and Linyphiidae were the most affected groups. The level of effect on individual species will be a function of many interacting factors (Jepson 1989). Application in the autumn, to seedling cereals, will not afford protection to diurnal species (Cilgi and Jepson, in press) and any refugia will result from being concealed in soil crevices. For those species active following spraying, the effects will be a function of residual exposure (Jepson *et al.* 1990a; Unal and Jepson 1991), susceptibility and physical factors such as temperature and humidity (Jepson *et al.* 1987; Croft 1990; Everts 1990). Many of these parameters remain unrecorded. It should also be pointed out that there is yet to be a study of population dynamics in fully treated fields over a series of generations, although experience would tend to indicate that effects can be intensified on this scale (Burn 1991; Aebischer 1989; Aebischer and Potts 1990; Jepson and Sherratt 1991). Previous long-term studies with pyrethroids have tended

to be within fields which incorporated unsprayed areas and thus provided opportunities for reinvasion over the period of observation (unpublished study reported in Inglesfield 1989).

In the present study, there was an apparent correlation between plot size and the duration of effects. This was most apparent with the Linyphiidae where the last significant effects at 1 and 4 ha were at 89 and 168 days, respectively. Inspection of the population trends for these families indicates that the effects may have extended for 150 and 230 days (1 and 4 ha, respectively). Reinvasion from control areas was not recorded, although it is known to be important for many of the predatory taxa that we studied (Jepson and Thacker 1990; Thomas *et al.* 1990). This would act to reduce the duration of recorded effects for all dispersive groups. Support for this argument arose from the barrier data in 1987/8 where densities of invertebrates in the barriered areas treated with deltamethrin, were consistently lower than within the open field areas that had received the same treatment. We assume that the plastic barrier will have acted in an equivalent way to a hedgerow in interrupting the reinvasion of treated areas from reservoir habitats and thus we postulate that whole-field treatments could be far more hazardous to the invertebrate population than partial treatments. This remains one of the most important problems for hazard evaluation of single products: it currently challenges the assumption of realism in this type of testing procedure and requires urgent investigation (Jepson 1992; Jepson and Sherratt 1991).

Despite their limitations as far as scale is concerned, small plots offer a beneficial compromise in terms of possible replication (Sotherton *et al.* 1988; Thacker 1991). The level of statistical separation of treatment effects was greater in 1987 than 1986. This should increase the probability of detecting small, but significant effects and also increase the duration over which effects can be detected on individual species, although neither of these was apparent in the present study. For the purposes of discriminating hazardous compounds, it may be better to strike the trade-off between replication and plot-size in favour of replication, using mathematical models, which take into account reinvasion, to predict the duration of effects on a realistic scale (Jepson 1992; Jepson *et al.* 1990b; Jepson and Sherratt 1991): over the time scale of recovery in the real world the nature of the pesticide will be less important and models will to a large extent concern the dispersal ability and population dynamics of the groups concerned. The use of small barriered areas within the treated field plots proved useful in 1987/8 and could become a formal requirement of testing protocols. It may also be advisable, when replication is sufficiently high, to employ the regular layout of plots used throughout this study, which avoided contiguous areas of the same treatment. This avoids heterogeneity in effective treatment areas, which could disrupt the analysis.

The data for *T. quadristriatus* did not reveal any indirect or sub-lethal effects. There was evidence that females were more active in the deltamethrin-treated plots; this could have been a result of increased foraging rates or an artefact of the preferential mortality of male beetles. This species was significantly depleted in the second year of the experiments. This is of interest given the apparent reversal of treatment effects for *T. quadristriatus* in the long-term studies of Burn (1991) and after summer treatment of winter-wheat, using organophosphorus aphicides, by Thacker (1991). These studies revealed an apparent resurgence which could be interpreted as a release from competition. That this did not occur after autumn treatment

may indicate that this is a sensitive phase in the life-cycle of *T. quadristriatus*, which is producing and laying eggs during the autumn and spring period. The long duration of this breeding period may offer some potential for phenological flexibility in this species (Mitchell 1963; Desender and Pollet 1987). However, the reasons for its variable response to pesticides remains unclear.

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